

Question 1

Define R_n to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For $i > 0$, let $p[i]$ denote the selling price of a rod whose length is i meters. Consider the array of prices:

$p[1]=1, p[2]=5, p[3]=8, p[4]=9, p[5]=10, p[6]=17, p[7]=18$

Which of the following statements is/are correct about R_7 ?

- ☐ $R_7=18$
- ☐ $R_7=19$
- ☐ R_7 is achieved by three different solutions
- ☐ R_7 cannot be achieved by a solution consisting of three pieces

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Question 2

For a C program accessing $X[i][j][k]$, the following intermediate code is generated by a compiler. Assume that the size of an integer is 32 bits and the size of a character is 8 bits.

```
t0 = i * 1024
t1 = j * 32
t2 = k * 4
t3 = t1 + t0
t4 = t3 + t2
t5 = X[t4]
```

Which one of the following statements about the source code for the C program is CORRECT?

- ☐ A X is declared as "int X[32][32][8]".
- ☐ B X is declared as "int X[4][1024][32]".
- ☐ C X is declared as "char X[4][32][8]".
- ☐ D X is declared as "char X[32][16][2]".

Question 3

A Young tableau is a 2D array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞ , and hence there cannot be any entry to the right of, or below a ∞ . The following Young tableau consists of unique entries.

1	2	5	14
3	4	6	23
10	12	18	25
31	∞	∞	∞

When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled in with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is _____

- ☐ A 2
- ☐ B 5
- ☐ C 6
- ☐ D 18

Question 4

Match each of the high level language statements given on the left hand side with the most natural addressing mode from those listed on the right hand side.

- | | |
|-------------------------------|-----------------------|
| 1 <code>A[1] = B[J];</code> | a Indirect addressing |
| 2 <code>while [*A++];</code> | b Indexed, addressing |
| 3 <code>int temp = *x;</code> | c Autoincrement |

☐ A (1, c), (2, b), (3, a)

☐ B (1, a), (2, c), (3, b)

☐ C (1, b), (2, c), (3, a)

☐ D (1, a), (2, b), (3, c)

Question 5

A program P reads in 500 integers in the range [0..100] representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for P to store the frequencies?

- ☐ A An array of 50 numbers
- ☐ B An array of 100 numbers
- ☐ C An array of 500 numbers
- ☐ D A dynamically allocated array of 550 numbers

Question 6

Consider the following declaration of a two dimensional array in C:

```
char a[100][100];
```



Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0, the address of a [40][50] is :

- ☐ A 4040
- ☐ B 4050
- ☐ C 5040
- ☐ D 5050

Question 7

Consider the following two C code segments. Y and X are one and two dimensional arrays of size n and $n \times n$ respectively, where $2 \leq n \leq 10$. Assume that in both code segments, elements of Y are initialized to 0 and each element $X[i][j]$ of array X is initialized to $i + j$. Further assume that when stored in main memory all elements of X are in same main memory page frame.

Code segment 1:

```
// initialize elements of Y to 0
// initialize elements X[i][j] of X to i+j
for (i = 0; i < n; i++)
    y[i] += X[0][i];
```

Code segment 2:

```
// initialize elements of Y to 0
// initialize elements X[i][j] of X to i+j
for (i = 0; i < n; i++)
    y[i] += X[i][0];
```

Which of the following statements is/are correct?

- S1: Final contents of array Y will be same in both code segments.
- S2: Elements of array X accessed inside the for loop shown in code segment 1 are contiguous in main memory.
- S3: Elements of array X accessed inside the for loop shown in code segment 2 are contiguous in main memory.

- ☐ A Only S2 is correct
- ☐ B Only S3 is correct
- ☐ C Only S1 and S2 are correct
- ☐ D Only S1 and S3 are correct

Question 8

Consider any array representation of an n element binary heap where the elements are stored from index 1 to index n of the array. For the element stored at index i of the array ($i \leq n$), the index of the parent is

- ☐ A $i - 1$
- ☐ B $\text{floor}(i/2)$
- ☐ C $\text{ceiling}(i/2)$
- ☐ D $(i+1)/2$

Question 9

In the 4×4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1 X 4 3

X 5 5 4

3 X 6 X

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is [GATE 2024 || Set-2 || 2 Marks || MCQ]

(A) 11

(B) 10

(C) 12

(D) 9

Question 10

Consider an array X that contains n positive integers. A subarray of X is defined to be a sequence of array locations with consecutive indices. The C code snippet given below has been written to compute the length of the longest subarray of X that contains at most two distinct integers. The code has two missing expressions labeled (P) and (Q).

Which one of the following options gives the CORRECT missing expressions?

(Hint: At the end of the i -th iteration, the value of $len1$ is the length of the longest subarray ending with $X[i]$ that contains all equal values, and $len2$ is the length of the longest subarray ending with $X[i]$ that contains at most two distinct values.)

```
int first=0, second=0, len1=0, len2=0, maxlen=0;
for (int i=0; i < n; i++) {
    if (X[i] == first) {
        len2++; len1++;
    } else if (X[i] == second) {
        len2++;
        len1 = (P) ;
        second = first;
    } else {
        len2 = (Q) ;
        len1 = 1;
        second = first;
    } if (len2 > maxlen) {
        maxlen = len2;
    } first = X[i];
}
```

(A) (P) $len1+1$ (Q) $len2+1$

(B) (P) 1 (Q) $len1+1$

(C) (P) 1 (Q) $len2+1$

(D) (P) $len2+1$ (Q) $len1+1$